EA-0027 - AQELYN Engineering Archive

AQELYN - EA-0027 Engineering Archive

IS-027 - AQELYN Identity Threat Detection & Behavioral Analytics Engine

Archive ID: EA-0027

Implementation Specification: IS-027

Component: AQELYN Identity Threat Detection & Behavioral Analytics Engine

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: COMPLETE

Repository Impact: No top-level repository structure changes

Breaking Changes: None

Predecessor Archives: EA-0001 through EA-0026

Next Specification: IS-028 - AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine

Document Control

Field	Value
Document	Engineering Archive EA-0027
Specification	IS-027 - AQELYN Identity Threat Detection & Behavioral Analytics Engine
Publication Format	Markdown, PDF, HTML, ZIP
Source of Truth	MD/EA-0027.md
Archive Rule	Implementation Specification -> Engineering Archive -> Continue
Repository Rule	Fixed repository structure; no redesign
Completion State	IS-027 complete; EA-0027 generated

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

Engineering Archive	Implementation Specification	Component
EA-0001	IS-001	AQELYN Kernel
EA-0002	IS-002	Universal Object Model
EA-0003	IS-003	AQELYN Event Bus
EA-0004	IS-004	AQELYN Evidence Engine

| EA-0005 | IS-005 | AQELYN Knowledge Graph |

| EA-0006 | IS-006 | AQELYN Trust Engine |

| EA-0007 | IS-007 | AQELYN Mission Engine |

| EA-0008 | IS-008 | AQELYN Workflow Engine |

| EA-0009 | IS-009 | AQELYN Policy Engine |

| EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |

| EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |

| EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |

| EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |

| EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |

| EA-0015 | IS-015 | AQELYN Security Operations Engine |

| EA-0016 | IS-016 | AQELYN Digital Forensics Engine |

| EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |

| EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |

| EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |

| EA-0020 | IS-020 | AQELYN AI Decision Intelligence Engine |

| EA-0021 | IS-021 | AQELYN Predictive Analytics & Forecasting Engine |

| EA-0022 | IS-022 | AQELYN Executive Intelligence & Strategic Reporting Engine |

| EA-0023 | IS-023 | AQELYN Threat Exposure & Attack Surface Management Engine |

| EA-0024 | IS-024 | AQELYN Vulnerability Intelligence & Prioritization Engine |

| EA-0025 | IS-025 | AQELYN Cyber Asset Discovery & Inventory Intelligence Engine |

| EA-0026 | IS-026 | AQELYN Configuration Compliance & Drift Intelligence Engine |

| EA-0027 | IS-027 | AQELYN Identity Threat Detection & Behavioral Analytics Engine |

The fixed repository structure remains:

```
AQELYN/
■■■ archive/
■■■ blueprint/
■■■ docs/
■■■ src/
■■■ tests/
■■■ tools/
■■■ build/
■■■ releases/
■■■ scripts/
■■■ assets/
■■■ examples/
■■■ plugins/
■■■ sdk/
■■■ api/
■■■ README.md
```

2. IS-027 Specification Identity

```
Specification ID: IS-027
Name: AQELYN Identity Threat Detection & Behavioral Analytics Engine
Engineering Archive Target: EA-0027
Project: AQELYN
System Type: Cyber Security Operating Environment
Status: Complete
Predecessor: IS-026 - AQELYN Configuration Compliance & Drift Intelligence Engine
```

3. Purpose

The AQELYN Identity Threat Detection & Behavioral Analytics Engine provides continuous identity monitoring, behavioral analytics, anomaly detection, insider threat identification, credential misuse detection, privilege abuse analysis, and identity risk intelligence across enterprise environments.

The engine establishes identity-centric threat detection by continuously analyzing authentication events, user behavior, privileged activities, service accounts, and machine identities using explainable AI and policy-governed analytics.

It answers:

```
Which identities exhibit anomalous behavior?
Which accounts are compromised or at risk?
Who is abusing privileged access?
Which authentication events indicate attacks?
Which identities require investigation?
Can every behavioral assessment be explained and audited?
```

4. Mission

The engine shall provide:

```
Identity behavioral analytics
Identity anomaly detection
Credential misuse detection
Privilege abuse detection
Identity risk scoring
Identity intelligence
Mission-aware identity reporting
Executive identity summaries
Continuous reassessment
Identity governance support
```

5. Scope

5.1 In Scope

```
Human identities
Machine identities
Service accounts
Privileged accounts
Cloud identities
Federated identities
Authentication events
Authorization events
```

5.2 Out of Scope

```
Password management
Identity provisioning
HR lifecycle management
Application source code
Physical access systems
```

6. Dependencies

IS-027 depends on:

```
IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
```

```
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 Identity & Access Governance Engine
IS-012 Asset & Configuration Governance Engine
IS-013 Risk Intelligence Engine
IS-014 Threat Intelligence Fusion Engine
IS-015 Security Operations Engine
IS-016 Digital Forensics Engine
IS-017 Threat Detection & Analytics Engine
IS-018 Automated Response & Orchestration Engine
IS-019 Security Data Lake & Telemetry Platform
IS-020 AI Decision Intelligence Engine
IS-021 Predictive Analytics & Forecasting Engine
IS-022 Executive Intelligence & Strategic Reporting Engine
IS-023 Threat Exposure & Attack Surface Management Engine
IS-024 Vulnerability Intelligence & Prioritization Engine
IS-025 Cyber Asset Discovery & Inventory Intelligence Engine
IS-026 Configuration Compliance & Drift Intelligence Engine
```

7. High-Level Architecture

```
AQELYN Identity Threat Detection & Behavioral Analytics Engine
■
■■■ Identity Behavior Engine
■■■ Anomaly Detection Engine
■■■ Credential Intelligence Engine
■■■ Privilege Analytics Engine
■■■ Identity Risk Engine
■■■ Behavioral Learning Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher
```

8. Functional Requirements

FR-027-001 - Identity Monitoring

The engine shall continuously monitor:

```
Authentication events
Authorization events
Privilege changes
Identity lifecycle events
Cloud identity activity
Machine identity activity
```

FR-027-002 - Behavioral Analytics

Continuously analyze:

```
User behavior
Authentication patterns
Access frequency
Session characteristics
Geographic anomalies
Behavioral baselines
```

FR-027-003 - Threat Detection

Detect:

```
Credential theft
Impossible travel
Privilege abuse
Insider threats
Account compromise
Identity anomalies
```

FR-027-004 - Explainable Identity Intelligence

Every assessment shall include:

```
Evidence references
Confidence indicators
Behavioral rationale
Risk explanation
Historical context
Policy references
```

FR-027-005 - Governance

Support:

```
Approval workflows
Policy validation
Version control
Auditability
Executive review
```

FR-027-006 - Event Publication

Publish standardized events:

```
identity.anomaly.detected
identity.risk.updated
credential.compromise.detected
privilege.abuse.detected
behavior.profile.updated
identity.investigation.created
```

9. Non-Functional Requirements

The engine shall provide:

```
Continuous monitoring
Enterprise scalability
Low-latency analytics
Explainability
Auditability
Repository stability
Backward compatibility
```

10. Core Identity Analytics Lifecycle

```
Identity Activity Collected
↓
Behavior Baseline
↓
Behavior Analysis
↓
Threat Detection
↓
Risk Assessment
↓
Policy Validation
↓
Continuous Learning
```

11. Internal Component Architecture

The AQELYN Identity Threat Detection & Behavioral Analytics Engine is implemented as a modular identity intelligence platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Identity Governance Engine, Risk Intelligence Engine, Threat Intelligence Fusion Engine, AI Decision Intelligence Engine, Executive Intelligence Engine, and Security Operations Engine.

```
AQELYN Identity Threat Detection & Behavioral Analytics Engine
■
■■■ Identity Behavior Engine
■■■ Anomaly Detection Engine
■■■ Credential Intelligence Engine
■■■ Privilege Analytics Engine
■■■ Identity Risk Engine
■■■ Behavioral Learning Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher
```

12. Component Specifications

12.1 Identity Behavior Engine

Continuously analyzes identity behavior.

Capabilities:

```
Authentication analysis
Authorization analysis
Session analytics
Behavior profiling
Activity correlation
```

12.2 Anomaly Detection Engine

Detects abnormal identity activity.

Supports:

```
Behavior deviation
Impossible travel
Unusual login patterns
Device anomalies
Access anomalies
```

12.3 Credential Intelligence Engine

Monitors credential-related threats.

Produces:

```
Credential compromise detection
Password spraying indicators
Credential stuffing indicators
Leaked credential correlation
Credential risk scoring
```

12.4 Privilege Analytics Engine

Analyzes privileged activity.

Produces:

```
Privilege escalation
Privilege abuse
Administrative anomalies
Role misuse
Permission changes
```

12.5 Identity Risk Engine

Calculates enterprise identity risk.

Supports:

```
Identity risk score
Behavior confidence
Threat severity
Mission impact
Business impact
```

12.6 Behavioral Learning Engine

Continuously improves behavioral models.

Supports:

```
Behavior baselines
Adaptive learning
False-positive reduction
Seasonality detection
Continuous optimization
```

13. Universal Object Model Extensions

13.1 IdentityBehaviorRecord

```
IdentityBehaviorRecord:
behavior_id
identity_id
confidence
risk_score
```

13.2 IdentityRiskAssessment

```
IdentityRiskAssessment:
assessment_id
severity
recommendation
generated_at
```

13.3 CredentialThreat

```
CredentialThreat:
threat_id
credential_status
threat_type
confidence
```

13.4 PrivilegeAssessment

```
PrivilegeAssessment:
assessment_id
privilege_level
anomaly_score
owner
```

14. Knowledge Graph Integration

Relationships:

```
Identity
↓
authenticates_to
↓
Asset

Identity
↓
assigned_role
↓
Privilege

Identity
↓
generates
↓
Behavior

Behavior
↓
creates
↓
Risk
```

15. Event Bus Integration

15.1 Identity Events

```
identity.login
identity.logout
identity.updated
```

15.2 Behavior Events

```
behavior.profile.updated
identity.anomaly.detected
```

15.3 Credential Events

```
credential.compromise.detected
credential.risk.updated
```

15.4 Privilege Events

```
privilege.abuse.detected
identity.risk.updated
identity.investigation.created
```

16. Security Data Lake Integration

Consumes:

```
Authentication telemetry
Authorization logs
Session history
Cloud identity telemetry
Identity audit logs
```

17. AI Decision Intelligence Integration

Consumes:

```
Behavior models
Risk scores
Investigation recommendations
Adaptive learning metrics
Confidence indicators
```

18. Risk & Threat Integration

Consumes:

```
Threat intelligence
Identity risk
Mission impact
Business criticality
Credential intelligence
```

19. Compliance Integration

Supports:

```
Identity governance
Access compliance
Policy validation
Audit evidence
Executive reporting
```

20. Public APIs

20.1 Identity API

```
GET /identities
GET /identities/{id}
POST /identities
```

20.2 Behavior API

```
GET /behavior
GET /behavior/{id}
```

20.3 Risk API

```
GET /identity-risk
GET /identity-risk/{id}
```

20.4 Investigation API

```
POST /investigations
GET /investigations/{id}
```

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
■■■ src/
■ ■■■ identity_behavior/
■■■ tests/
■ ■■■ identity_behavior/
■■■ docs/
■ ■■■ identity_behavior/
■■■ api/
■ ■■■ identity_behavior/
■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Identity Threat Detection & Behavioral Analytics Engine is the trusted subsystem responsible for continuously monitoring identity behavior, detecting anomalies, assessing identity risk, identifying credential compromise, and governing identity threat intelligence.

Every identity assessment shall be:

```
Explainable
Evidence-backed
Policy-governed
Behavior-aware
Risk-aware
Mission-aware
Fully auditable
Continuously reassessed
```

22.1 Security Principles

```
Zero Trust
Defense in Depth
Least Privilege
Continuous Identity Monitoring
Secure by Design
Policy Enforcement
Explainable Identity Intelligence
Continuous Learning
```

22.2 Authorization Model

Supported operational roles:

```
Identity Administrator
SOC Analyst
Threat Hunter
Incident Responder
Security Administrator
Mission Owner
Compliance Officer
Executive Reviewer
```

All identity investigations and behavioral assessments shall be governed through the AQELYN Policy Engine.

22.3 Identity Assessment Integrity

Identity assessments shall maintain:

```
Unique assessment identifier
Identity reference
Behavior baseline reference
Evidence references
Risk score
Confidence indicators
Recommendation
Audit trail
```

Assessment history shall be append-only.

22.4 Behavioral Model Integrity

Behavioral models shall maintain:

```
Model identifier
Baseline window
Learning version
Training evidence
Confidence metrics
False-positive feedback
Policy references
Audit record
```

No behavioral model shall be considered authoritative without version metadata, evidence lineage, and validation state.

23. Identity Analytics Lifecycle

23.1 Behavioral Assessment Lifecycle

```
Identity Activity Collected
↓
Behavior Baseline
↓
Behavior Analysis
↓
Threat Detection
↓
Risk Assessment
↓
Continuous Learning
```

23.2 Investigation Lifecycle

```
Threat Detected
↓
Investigation Created
↓
Evidence Correlated
↓
Risk Confirmed
↓
Response Initiated
↓
Case Closed
```

23.3 Audit Lifecycle

```
Assessment Created
↓
Evidence Linked
↓
Policy Validated
```

```
↓
Audit Stored
```

24. Continuous Identity Operations

The engine continuously evaluates:

```
Authentication activity
Authorization activity
Credential usage
Privilege changes
Behavior anomalies
Identity risk
Machine identities
Cloud identities
```

25. Performance Requirements

The engine shall support:

```
Continuous monitoring
Low-latency behavioral analytics
Enterprise-scale identity analysis
Concurrent investigations
High availability
Continuous operation
```

26. Scalability Requirements

The engine shall scale to support:

```
Global enterprises
Hybrid identity environments
Multi-cloud identity providers
Millions of identities
Distributed analytics
Long-term behavioral history
```

27. Audit Requirements

Every identity operation shall generate immutable audit records.

Audit events include:

```
Identity authenticated
Behavior profile updated
Credential compromise detected
Privilege abuse detected
Identity risk updated
Investigation completed
```

28. Failure Handling

28.1 Behavioral Analysis Failure

```
Analysis retried
Failure recorded
Administrator notified
```

28.2 Threat Detection Failure

```
Detection retried
Previous assessment retained
Audit generated
```

28.3 Learning Failure

```
Learning model rolled back
Manual review initiated
Audit recorded
```

28.4 Policy Failure

```
Investigation blocked
Policy violation recorded
Manual approval required
```

29. Testing Strategy

29.1 Unit Testing

Validate:

```
Identity Behavior Engine
Anomaly Detection Engine
Credential Intelligence Engine
Privilege Analytics Engine
Identity Risk Engine
Behavioral Learning Engine
```

29.2 Integration Testing

Verify interaction with:

```
Kernel
Knowledge Graph
Security Data Lake
Identity Governance Engine
Threat Intelligence Engine
Risk Intelligence Engine
AI Decision Engine
Executive Reporting Engine
Security Operations Engine
```

29.3 System Testing

Validate:

```
Behavior analytics
Anomaly detection
Credential intelligence
Privilege analytics
Identity investigations
Audit generation
```

29.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Explainability
Audit logging
Identity assessment integrity
```

29.5 Regression Testing

Verify IS-001 through IS-026 remain unaffected.

30. Acceptance Criteria

IS-027 is complete when:

```
Identity Behavior Engine implemented
Anomaly Detection Engine implemented
Credential Intelligence Engine implemented
Privilege Analytics Engine implemented
Identity Risk Engine implemented
Repository unchanged
Testing documented
```

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/identity_behavior/
■■■ tests/identity_behavior/
■■■ docs/identity_behavior/
■■■ api/identity_behavior/
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-027 introduces the AQELYN Identity Threat Detection & Behavioral Analytics Engine, providing enterprise-scale identity monitoring, behavioral analytics, anomaly detection, credential intelligence, privilege abuse detection, explainable identity risk scoring, and continuous identity learning.

Major capabilities include:

```
Identity Behavioral Analytics
Identity Anomaly Detection
Credential Intelligence
Privilege Abuse Detection
Identity Risk Scoring
Behavioral Learning
Mission-Aware Identity Intelligence
Executive Identity Reporting
Continuous Identity Monitoring
Identity Governance
```

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

```
Specification ID : IS-027
Title : AQELYN Identity Threat Detection & Behavioral Analytics Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0027
```

34. EA-0027 Engineering Objective

The objective of IS-027 was to introduce a dedicated Identity Threat Detection & Behavioral Analytics Engine that enables AQELYN to continuously monitor identities, establish behavioral baselines, detect anomalies, identify credential misuse, detect privilege abuse, assess identity risk, and create identity investigations.

The engine extends AQELYN from configuration compliance into identity-centric detection and behavioral intelligence.

35. EA-0027 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Identity monitoring
Behavioral analytics
Identity anomaly detection
Credential intelligence
Privilege analytics
Identity risk scoring
Behavioral learning
Knowledge Graph integration
Security Data Lake integration
Risk and threat intelligence integration
AI Decision integration
Executive reporting integration
Event publishing
```

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated Identity Threat Detection Engine

Identity threat detection and behavioral analytics are implemented as a standalone engine rather than embedded in Identity Governance or Security Operations.

Rationale:

```
Clear separation of identity administration from identity threat analytics.
Independent lifecycle and governance.
Better support for behavioral baselines, anomaly detection, and identity risk scoring.
Improved traceability for identity investigations.
```

36.2 Decision 2 - Behavior Is Baseline-Driven

Behavioral analytics compare current activity against learned baselines, session characteristics, geography, access frequency, and identity context.

Benefits:

```
Anomalies can be identified even without known indicators.
Identity risk becomes adaptive.
```

```
False positives can be reduced through behavioral learning.
```

36.3 Decision 3 - Credential and Privilege Analytics Are First-Class Capabilities

Credential threats and privilege abuse are modeled directly rather than treated as generic alerts.

Benefits:

```
Credential compromise receives specific detection logic.
Privileged misuse can be prioritized.
Identity investigations can distinguish account compromise from role misuse.
```

36.4 Decision 4 - Identity Assessments Are Evidence-Backed

Every identity assessment includes evidence references, confidence, behavioral rationale, risk explanation, historical context, and policy references.

Benefits:

```
Identity decisions become defensible.
Security teams can explain investigations.
Compliance teams can audit identity risk decisions.
```

36.5 Decision 5 - Event-Driven Identity Analytics Lifecycle

Identity login, behavior profile, credential, privilege, risk, and investigation events are published through the AQELYN Event Bus.

Examples include:

```
identity.login
identity.logout
identity.updated
behavior.profile.updated
identity.anomaly.detected
credential.compromise.detected
credential.risk.updated
privilege.abuse.detected
identity.risk.updated
identity.investigation.created
```

36.6 Decision 6 - Universal Object Model Extension

New domain objects introduced include:

```
IdentityBehaviorRecord
IdentityRiskAssessment
CredentialThreat
PrivilegeAssessment
```

37. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Behavior, identity risk, credential threat, privilege assessment objects |

| IS-003 Event Bus | Identity, behavior, credential, privilege, risk events |

| IS-004 Evidence Engine | Evidence references and identity support |

| IS-005 Knowledge Graph | Identity, asset, privilege, behavior, risk relationships |

| IS-006 Trust Engine | Data confidence and identity trust |

| IS-007 Mission Engine | Mission-aware identity risk prioritization |

| IS-008 Workflow Engine | Investigation and response workflows |

| IS-009 Policy Engine | Identity governance and investigation policies |

| IS-010 Compliance Engine | Identity audit and access compliance |

| IS-011 Identity Governance Engine | Identity definitions, roles, lifecycle, entitlements |

| IS-013 Risk Intelligence Engine | Risk scoring and business impact |

| IS-014 Threat Intelligence Engine | Credential and threat actor context |

| IS-015 Security Operations Engine | Investigations and SOC workflow |

| IS-019 Security Data Lake | Authentication telemetry and identity logs |

| IS-020 AI Decision Engine | Investigation recommendations and learning insights |

| IS-022 Executive Reporting Engine | Executive identity summaries |

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/identity_behavior/
■■■ tests/identity_behavior/
■■■ api/identity_behavior/
■■■ docs/identity_behavior/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces identity-threat-specific security controls:

```
Policy-governed identity investigations
Evidence-backed behavioral assessments
Continuous identity monitoring
Credential compromise detection
Privilege abuse detection
Identity risk scoring
Behavioral model audit trail
Identity investigation traceability
Role-authorized identity analytics administration
```

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

```
Identity behavioral analytics
Identity anomaly detection
Credential misuse detection
Privilege abuse detection
Identity risk scoring
```

```
Identity intelligence
Mission-aware identity reporting
Executive identity summaries
Continuous reassessment
Identity governance support
```

41. Risks Identified

Risk	Mitigation
False-positive identity anomalies	Behavioral learning and confidence indicators
Missed credential compromise	Credential Intelligence Engine and threat correlation
Privilege abuse blind spots	Privilege Analytics Engine and audit logging
Unauthorized investigations	Policy enforcement and role authorization
Biased behavioral models	Learning audit, evidence lineage, review workflows
Incomplete identity telemetry	Security Data Lake integration and source tracking
Privacy concerns	Least privilege, audit, policy governance
Excessive alerting	Risk scoring and investigation prioritization

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

```
Unit testing
Integration testing
System testing
Security testing
Regression testing
```

Acceptance criteria cover identity behavior, anomaly detection, credential intelligence, privilege analytics, identity risk, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

```
Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline
```

44. Dependencies

Required:

```
EA-0001 through EA-0026
IS-001 through IS-027
```

Enables:

```
IS-028 and subsequent cloud posture, SaaS posture, supply chain intelligence, and cyber resilience
components
```

45. Completion Record

```
Engineering Archive : EA-0027
Implementation Specification : IS-027
Title : AQELYN Identity Threat Detection & Behavioral Analytics Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
```

46. Archive Index Update

```
EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
EA-0019 IS-019 AQELYN Security Data Lake & Telemetry Platform
EA-0020 IS-020 AQELYN AI Decision Intelligence Engine
EA-0021 IS-021 AQELYN Predictive Analytics & Forecasting Engine
EA-0022 IS-022 AQELYN Executive Intelligence & Strategic Reporting Engine
EA-0023 IS-023 AQELYN Threat Exposure & Attack Surface Management Engine
EA-0024 IS-024 AQELYN Vulnerability Intelligence & Prioritization Engine
EA-0025 IS-025 AQELYN Cyber Asset Discovery & Inventory Intelligence Engine
EA-0026 IS-026 AQELYN Configuration Compliance & Drift Intelligence Engine
EA-0027 IS-027 AQELYN Identity Threat Detection & Behavioral Analytics Engine
```

47. Engineering Phase Status

```
Completed Engineering Archives : EA-0001 through EA-0027

Current Status:
EA-0027 COMPLETE

Next Implementation Specification:
IS-028 - AQELYN Cloud Security Posture Management (CSPM) Intelligence Engine
```

48. Engineering Archive Publication Standard

EA-0027 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg
■ ■■■ Component.svg
■ ■■■ Workflow.svg
■ ■■■ EventFlow.svg
■ ■■■ Integration.svg
■
■■■ examples/
■ ■■■ example_*.md
■
■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
■
■■■ images/
■
■■■ journal/
■ ■■■ Engineering_Journal.md
■
■■■ manifest/
■ ■■■ manifest.json
■
■■■ MD/
■ ■■■ EA-xxxx.md
■
■■■ PDF/
■ ■■■ EA-xxxx.pdf
■
■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-027-001	Continuously monitor identities	Sections 8, 12, 23	Complete
FR-027-002	Perform behavioral analytics	Sections 8, 12, 23	Complete
FR-027-003	Detect identity threats	Sections 8, 12, 23	Complete
FR-027-004	Provide explainable identity intelligence	Sections 8, 22, 36	Complete
FR-027-005	Support governance	Sections 8, 22, 23	Complete
FR-027-006	Publish identity events	Sections 8, 15, 36	Complete
NFR-027-001	Continuous monitoring	Sections 9, 24	Complete
NFR-027-002	Enterprise scalability	Sections 9, 25, 26	Complete
NFR-027-003	Low-latency analytics	Sections 9, 25	Complete
NFR-027-004	Explainability	Sections 9, 22, 36	Complete
NFR-027-005	Auditability	Sections 9, 27, 39	Complete
NFR-027-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

Source	Target	Relationship
IS-027 Purpose	EA-0027 Objective	Defines why the engine exists
Identity Behavior Engine	FR-027-001, FR-027-002	Monitors and analyzes behavior
Anomaly Detection Engine	FR-027-003	Detects behavioral anomalies
Credential Intelligence Engine	FR-027-003	Detects credential threats
Privilege Analytics Engine	FR-027-003	Detects privilege abuse
Identity Risk Engine	Risk assessment	Calculates identity risk
Behavioral Learning Engine	Continuous learning	Improves behavior models
Event Publisher	FR-027-006	Publishes identity events
Security Data Lake Integration	Identity telemetry	Supplies authentication and session data
AI Decision Integration	Investigation recommendations	Supplies confidence and recommendations
Risk & Threat Integration	Identity risk context	Supplies threat and risk context
Compliance Integration	Identity governance	Supports access compliance and audit
Repository Validation	Repository Standard	Confirms no top-level redesign

51. Engineering Journal

Journal Entry - EA-0027

EA-0027 was created to archive completion of IS-027 - AQELYN Identity Threat Detection & Behavioral Analytics Engine.

The archive records the expansion of AQELYN into identity threat detection and behavioral analytics. IS-027 defines the structure needed to monitor identity activity, establish behavior baselines, detect identity anomalies, identify credential threats, analyze privilege abuse, calculate identity risk, create investigations, and publish identity threat events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Identity threat detection must be modeled separately from identity governance. Identity governance defines roles and access; identity threat detection analyzes behavior, compromise, abuse, and risk.

Governance Note

EA-0027 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Identity Behavior Record

```
behavior_id: BEH-0001
identity_id: ID-1001
confidence: 0.91
risk_score: 82
behavior_summary: unusual privileged access from new geography
```

52.2 Example Identity Risk Assessment

```
assessment_id: IDRISK-2001
severity: high
recommendation: create investigation and require step-up verification
generated_at: 2026-07-07T12:00:00Z
```

52.3 Example Credential Threat

```
threat_id: CRED-3001
credential_status: suspected_compromised
threat_type: password_spraying
confidence: 0.87
```

52.4 Example Identity Event

```
{
"event_type": "identity.anomaly.detected",
"identity_id": "ID-1001",
"risk_score": 82,
"source_engine": "aqelyn_identity_threat_detection_behavioral_analytics_engine"
}
```

53. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0027.md
PDF/EA-0027.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_identity_behavior.md
```

54. Final Archive Statement

EA-0027 is the Engineering Archive for IS-027 - AQELYN Identity Threat Detection & Behavioral Analytics Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0027 COMPLETE
IS-027 COMPLETE
NEXT: IS-028
```